
Probability and Statistic (for CS Students)

Lecture 6: Conditional Probability and Bayes II

University of Mazandaran

Fall 1405

Conditional probability of E given F:

$$P(E|F) = \frac{P(E, F)}{P(F)} \quad \longleftarrow \text{Main Definition}$$

$$P(E, f) = P(E|F) \times P(F) \quad \longleftarrow \text{Chain Rule}$$

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General form of chain rule:

$$P(E_1, E_2, \dots, E_n) = P(E_1) P(E_2|E_1) P(E_3|E_1, E_2) \cdots P(E_n|E_1, E_2, \dots, E_{n-1})$$

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Law of total probability:

$$P(E) = P(E|F) P(F) + P(E|F^c) P(F^c)$$

Evolution of Bacteria

You have bacteria in your gut which is causing a disease.

10% have a mutation which makes them resistant to anti-biotics.

You take half a course of anti-biotics ...

Probability a bacteria survives given it has the mutation: 20%

Probability a bacteria survives given it doesn't have the mutation:

1%

What is the probability that a randomly chosen bacteria survives?

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S : the event that a bacteria survives

M : the event that a bacteria has the mutation

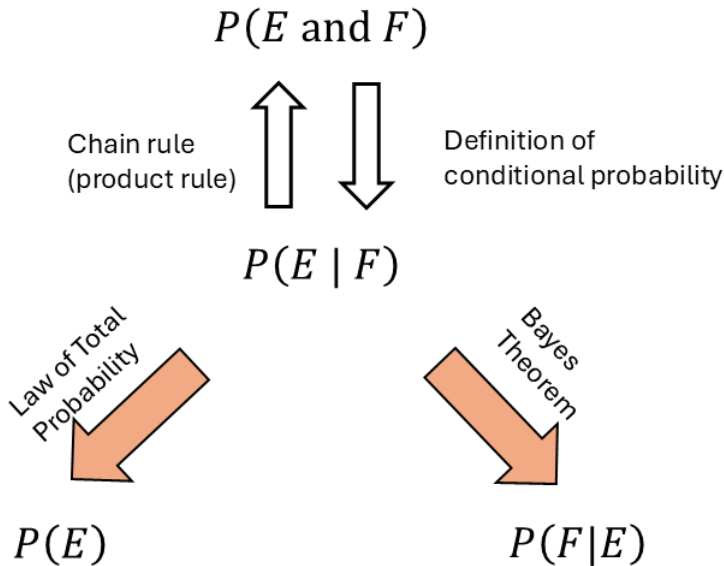
$$\begin{aligned}P(S) &= P(S \text{ and } M) + P(S \text{ and } M^c) \\&= P(S|M)P(M) + P(S|M^c)P(M^c) \\&= 0.2 \times 0.1 + 0.01 \times 0.9 = 0.029\end{aligned}$$

Real question: what is the probability that a surviving bacteria has the mutation?

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$$P(M|S)$$

Relationship Between Probabilities



Bayes' Theorem



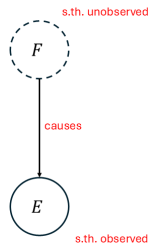
I want to calculate
 $P(\text{State of the world } F | \text{Observation } E)$
It seems so tricky!...

The other way around is easy
 $P(\text{Observation } E | \text{State of the world } F)$.
What options do I have?



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Want $P(F|E)$; Know $P(E|F)$

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A little while later ...

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$$= \frac{P(E|F)P(F)}{P(E|F)P(F) + P(E|F^c)P(F^c)}$$

Bayes' Theorem

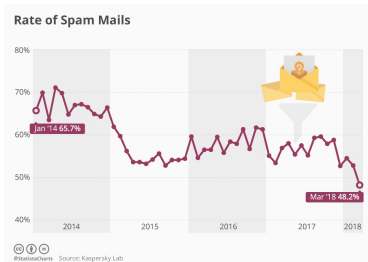
Theorem: For any events E and F where $P(E) > 0$ and $P(F) > 0$,

$$P(F|E) = \frac{P(E|F)P(F)}{P(E)}$$

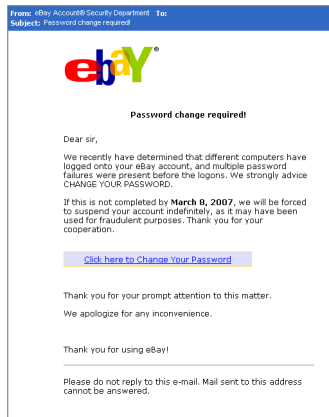
Expanded form:

$$P(F|E) = \frac{P(E|F)P(F)}{P(E|F)P(F) + P(E|F^c)P(F^c)}$$

Detecting spam email



We can easily calculate how many
spam emails contain “Dear”:
 $P(E|F) = P(\text{“Dear”} | \text{Spam email})$



But, what is the probability that an
email containing “Dear” is spam?
 $P(F|E) = P(\text{Spam email} | \text{“Dear”})$

Detecting spam email

You get an email with the word “Dear” in it.
What is the probability that the email is spam?

- ▶ 60% of all emails are spam,
- ▶ 20% of spam has the word “Dear”
- ▶ 1% of non-spam has the word “Dear”

Detecting spam email

1. Define events & state goal
2. Identify known probabilities
3. Solve

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Let: E : "Dear"; F : spam

Want: $P(\text{spam} | \text{"Dear"})$

$$= P(F|E)$$

Identify known probabilities

- ▶ 60% of all email in 2016 is spam,

$$P(F|E) = \frac{P(E|F) P(F)}{P(E|F) P(F) + P(E|F^c) P(F^c)}$$

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Solve

- ▶ 60% of all email in 2016 is spam,
- ▶ 20% of spam has the word “Dear”
- ▶ 1% of non-spam has the word “Dear”

$$P(F|E) = \frac{.2 \times .6}{.2 \times .6 + .1 \times .4}$$

$$P(F|E) = \frac{P(E|F)P(F)}{P(E|F)P(F) + P(E|F^c)P(F^c)}$$

$$P(F|E) = \frac{P(E|F)P(F)}{P(E)}$$

- ▶ likelihood: $P(E|F)$
- ▶ prior: $P(F)$
- ▶ evidence: $P(E)$
- ▶ posterior: $P(F|E)$

SARS Virus Testing

A test is 98% effective at detecting SARS

- ▶ However, test has a “false positive” rate of 1%
- ▶ 0.5% of US population has SARS
- ▶ Let E = you test positive for SARS with this test
- ▶ Let F = you actually have SARS
- ▶ What is $P(F|E)$

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Solution:

$$P(F|E) = \frac{P(E|F)P(F)}{P(E|F)P(F) + P(E|F^c)P(F^c)}$$

$$P(F|E) = \frac{0.98 \times 0.005}{0.98 \times 0.005 + 0.01 \times (1 - 0.005)} \approx 0.330$$

Bayes Theorem Intuition



Bayes Theorem Intuition

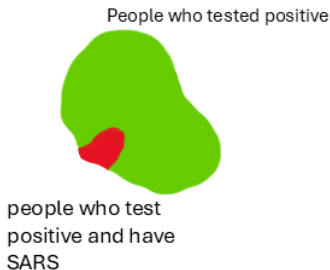


Bayes Theorem Intuition



Bayes Theorem Intuition

Conditioning on a positive result changes the sample space to this:



Multiple Choice Theory

Consider a multiple choice midterm question:

Let $3/4$ be the probability that a learner knows the concepts to the midterm question.

Let $1/4$ be the probability that a learner gets the answer correct if they **don't know** the concepts.

Let $1/10$ be the probability that a learner gets the question incorrect given they **do know** the concepts.

What is the probability they know the concept, given they answered correct?

1. Define events & state the goal

K : the event that the learner **Knows** the concept

C : the event that the learner gets the answer **Correct**

2. Identify known probabilities

$$P(K) = \frac{3}{4}$$

$$P(C|K') = \frac{1}{4}$$

$$P(C'|K) = \frac{1}{10}$$

$$P(K|C) = ?$$

3. Solve

$$P(K|C) = \frac{P(K,C)}{P(C)} = \frac{P(C|K)P(K)}{P(C|K)P(K) + P(C|K')P(K')}$$

3. Solve

$$\begin{aligned} P(K|C) &= \frac{P(K,C)}{P(C)} = \frac{P(C|K)P(K)}{P(C|K)P(K)+P(C|K')P(K')} \\ &= \frac{0.9 \times 0.75}{(0.9 \times 0.75) + (0.25 \times 0.25)} \end{aligned}$$

$= 1 - P(C'|K)$



Let's play a game!

5 Hand Poker!

Scenario:

- ▶ You play against an opponent
- ▶ A standard deck of 52 cards is used
- ▶ Each player receives 5 private cards
- ▶ The player with the stronger poker hand wins

Before making your decision, you observe your opponent's behavior

A **tell** is an observable behavior that may reveal information about an opponent's hand

Your opponent gets excited if they are winning ...

probability of a tell given they **have a winning** hand: **0.5**

probability of a tell given they **don't have a winning** hand: **0.1**

Telling in Cards

You are playing a game. Your opponent has two unseen cards. If either is an ace, you lose ...

The following cards are seen by you (they are not in your opponent's hand):

2D 6D 5H 6S JH 2H

5D 3H KD 10C 10S QS

9C JD QC 5S 8H 7C 10H

Opponent does **not** have a tell ...

Do you choose to play? yes or no



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